Components of Physical Fitness and the FITT Principle

This is my reviewer for our Sports track lesson on fitness. Physical fitness is the body's ability to do daily tasks and physical activities with energy left over for fun and emergencies. It is split into health-related and skill-related components. After that, we plan workouts using the FITT principle.

Health-Related Components

These five affect your overall health and lower your risk of lifestyle diseases. Everyone needs them, athlete or not.

- Cardiovascular endurance: the ability of the heart, lungs, and blood vessels to supply oxygen to working muscles during long activity. Jogging around the barangay, swimming, and Zumba build it. The 3-minute step test is a common check.
- Muscular strength: the greatest force a muscle can produce in one effort. Lifting a heavy sack of rice once is a strength task.
- Muscular endurance: the ability of a muscle to repeat a movement many times, or hold a position, without tiring. Many push-ups in a row or a long plank test it.
- Flexibility: the range of motion around a joint. The sit-and-reach test checks the lower back and hamstrings.
- Body composition: the proportion of fat mass to lean mass like muscle, bone, and water. Body mass index, or BMI, is a simple screening estimate, but it does not directly measure fat.

Tip: strength is one big push; endurance is many smaller pushes.

Skill-Related Components

These six help you perform in sports. They depend partly on training and partly on natural ability.

- Agility: changing direction quickly while keeping control, like a basketball player cutting past a defender. The hexagon agility test checks it.
- Balance: keeping the body steady while still (static balance) or while moving (dynamic balance). Standing on one foot is static; walking on a beam is dynamic.
- Coordination: making body parts work together smoothly, often with the eyes. Kicking a sepak takraw ball needs eye-foot coordination.
- Power: producing strength quickly, so it combines strength and speed. A volleyball spike jump or a standing long jump shows power.
- Reaction time: the time between noticing a signal and starting to respond, like a sprinter reacting to the starting gun.
- Speed: moving the body or a body part from one point to another in the shortest time. The 40-meter sprint is a typical test.

Do not mix up the last two. Reaction time is how fast you start; speed is how fast you move once started.

The FITT Principle

FITT is a guide for planning exercise. Each letter is a variable you can adjust.

- Frequency: how often you exercise, usually sessions per week, like jogging three times a week.

- Intensity: how hard you work, measured by heart rate, weight lifted, or how hard the effort feels.
- Time: how long each session lasts, such as 30 minutes of brisk walking.
- Type: the kind of activity, such as running, biking, weight training, or yoga. Match the type to the component you want to improve.

Training Principles

These explain how the body adapts to exercise.

- Overload: to improve, you must challenge the body more than it is used to. If you always do 10 squats, your legs eventually stop improving.
- Progression: increase the load gradually, not all at once, to reduce injury risk. Many coaches suggest raising weekly volume by no more than about ten percent.
- Specificity: you get better at what you train. Training only your legs will not do much for badminton smashes.
- Reversibility: fitness gains are slowly lost when you stop training, the "use it or lose it" idea. After a long vacation you may tire faster.

Warm-Up and Cool-Down

A warm-up prepares the body for harder work. It usually lasts 5 to 10 minutes, with light cardio like brisk walking plus dynamic stretches such as arm circles and leg swings. It gradually raises body temperature and heart rate and helps lower injury risk.

A cool-down brings heart rate and breathing back to normal after exercise. It usually includes slow walking, then static stretches held for about 15 to 30 seconds. Skipping it after intense work can leave you feeling dizzy.

Target Heart Rate

To check cardio intensity, we use heart rate. First, estimate your maximum heart rate with 220 minus your age. For a 17-year-old, that is 220 minus 17, or 203 beats per minute.

Your target zone is a percentage of that maximum. Moderate activity is about 50 to 70 percent; vigorous activity is about 70 to 85 percent. So the 17-year-old's moderate zone is roughly 102 to 142 beats per minute.

Check your pulse at the wrist or the side of the neck. Count beats for 15 seconds and multiply by 4. The formula is only an estimate, and anyone with a heart condition should follow a doctor's advice.

Quick check

- Q: Which health-related component is tested by the sit-and-reach? A: Flexibility.
- Q: Which skill-related component combines strength and speed? A: Power.
- Q: What does the second T in FITT stand for? A: Type.
- Q: Which training principle says gains are lost when you stop exercising? A: Reversibility.
- Q: What is the estimated maximum heart rate of a 20-year-old? A: 200 beats per minute, from 220 minus 20.